CO1 – Assessment Test 3 (AT3)

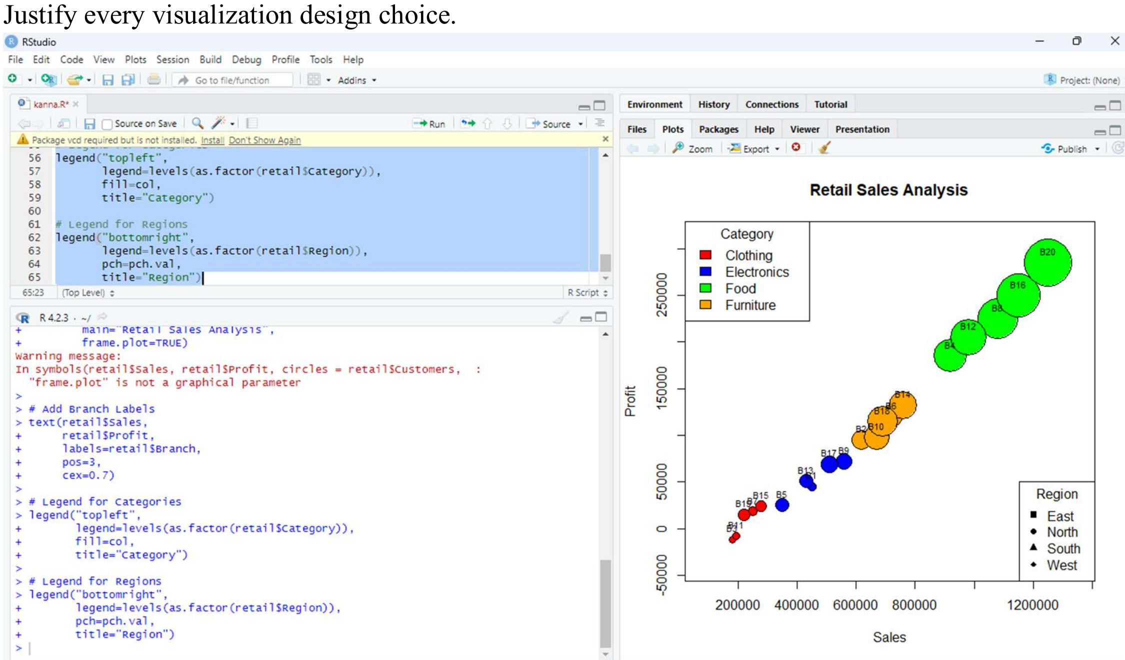
Visualization Design Exercise

s Done by: Roshan R(192524085)

Question 1

Question

A multinational retail company has collected annual sales information from its branches across different regions. Using the given dataset, design an effective visualization that simultaneously represents Sales, Profit, Number of Customers, Product Category, and Region. Select suitable aesthetic mappings and apply an appropriate scale transformation wherever necessary. Justify every visualization design choice.



R-Program:

```
# Create Dataset
retail <- data.frame(
Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",
"B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),

Region=c("North","South","East","West","North","South","East","West",
"North","South","East","West","North","South","East","West",
"North","South","East","West"),
```

```
Category=c("Electronics","Furniture","Clothing","Food","Electronics",
"Furniture","Clothing","Food","Electronics","Furniture",
"Clothing","Food","Electronics","Furniture","Clothing",
"Food","Electronics","Furniture","Clothing","Food"),

Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,
560000,670000,195000,980000,430000,760000,280000,1150000,
510000,690000,220000,1250000),

Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
72000,98000,-8000,205000,51000,132000,24000,250000,
69000,115000,15000,285000),

Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
2100,3300,1100,4700,1750,3600,1450,5600,
2200,3900,1600,6100)
)

# Assign Colors for Categories
col <- c("red","blue","green","orange")
cat.col <- col[as.factor(retail$Category)]

# Assign Symbols for Regions
pch.val <- c(15,16,17,18)
reg.sym <- pch.val[as.factor(retail$Region)]

# Bubble Plot
symbols(retail$Sales,
        retail$Profit,
        circles=retail$Customers,
        inches=0.3,
        bg=cat.col,
        fg="black",
        xlab="Sales",
        ylab="Profit",
        main="Retail Sales Analysis",
        frame.plot=TRUE)

# Add Branch Labels
text(retail$Sales,
     retail$Profit,
     labels=retail$Branch,
     pos=3,
     cex=0.7)

# Legend for Categories
legend("topleft",
       legend=levels(as.factor(retail$Category)),
       fill=col,
```

```
    title="Category")

# Legend for Regions
legend("bottomright",
    legend=levels(as.factor(retail$Region)),
    pch=pch.val,
    title="Region")
```

Explanation:

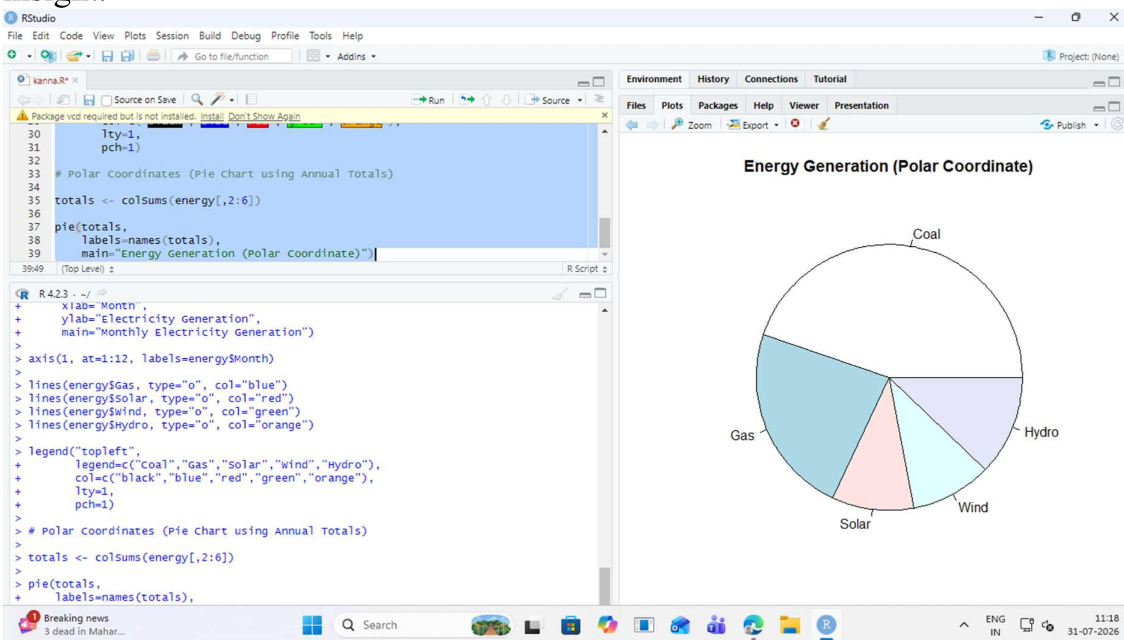
The retail dataset containing Branch, Region, Category, Sales, Profit, and Customers is created in R. Different colors are assigned to product categories and different symbols are assigned to regions. A bubble plot is generated using Sales on the X-axis and Profit on the Y-axis. The size of each bubble represents the number of customers, and branch names are displayed above each bubble. Legends are added to identify product categories and regions, making the visualization easy to interpret.

Justification

- **Sales** → X-axis
- **Profit** → Y-axis
- **Customers** → Bubble size
- **Category** → Bubble color
- **Region** → Different plotting symbols
- **Branch** → Labels

This visualization displays all five variables in a single graph, making it easy to compare branch performance across different regions and product categories. It satisfies all the requirements of the question.

Question 2
Question
The following dataset contains monthly electricity generation from different energy sources.
Develop visualizations using Cartesian Coordinates and Polar Coordinates. Compare both
visualizations and determine which coordinate system provides better analytical insight.



R-program:

Create Dataset

energy <- data.frame(

Month = month.abb,

Coal = c(620,580,590,605,640,670,690,720,700,680,650,630),

Gas = c(310,295,305,320,340,355,360,370,365,350,335,320),

Solar = c(45,60,90,130,180,230,250,245,210,160,90,50),

Wind = c(80,95,120,150,170,185,195,190,175,150,110,90),

Hydro = c(140,150,160,170,180,195,205,210,200,185,165,150)

)

```
# Cartesian Coordinates (Line Graph)

plot(energy$Coal, type="o", col="black",
     ylim=c(0,750),
     xaxt="n",
     xlab="Month",
     ylab="Electricity Generation",
     main="Monthly Electricity Generation")

axis(1, at=1:12, labels=energy$Month)

lines(energy$Gas, type="o", col="blue")
lines(energy$Solar, type="o", col="red")
lines(energy$Wind, type="o", col="green")
lines(energy$Hydro, type="o", col="orange")

legend("topleft",
       legend=c("Coal","Gas","Solar","Wind","Hydro"),
       col=c("black","blue","red","green","orange"),
       lty=1,
       pch=1)

# Polar Coordinates (Pie Chart using Annual Totals)
```

```
totals <- colSums(energy[,2:6])

pie(totals,
    labels=names(totals),
    main="Energy Generation (Polar Coordinate)")
```

Explanation:

The monthly electricity generation dataset is created in R with five energy sources: Coal, Gas, Solar, Wind, and Hydro. A line graph is plotted using Cartesian coordinates to compare the monthly trend of electricity generation for each energy source. Different colors are used for easy identification, and a legend is added. The annual generation totals are then calculated and represented using a pie chart, which is a polar coordinate visualization. This helps compare the overall contribution of each energy source.

Comparison (Answer for the Question)

Cartesian Coordinates	Polar Coordinates
Shows monthly trends clearly.	Shows overall contribution of each energy source.
Easy to compare values across months.	Easy to compare proportions.
Best for analyzing increases and decreases.	Best for showing percentage share.

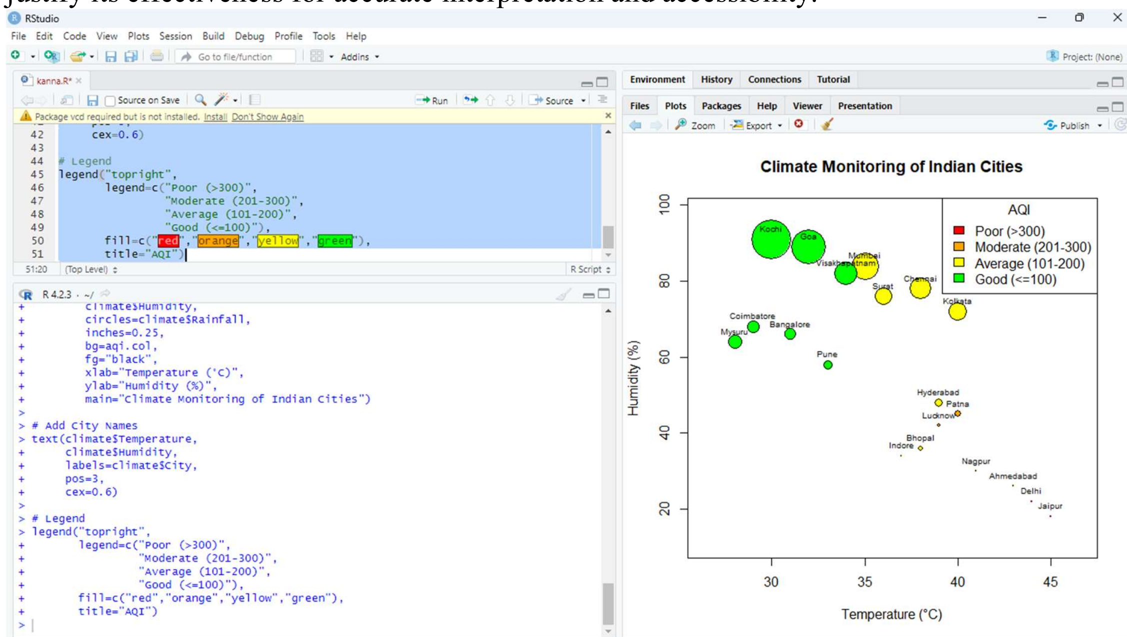
Conclusion

Cartesian coordinates provide better analytical insight because they clearly show the monthly variation and trends in electricity generation for each energy source. Polar coordinates are useful for understanding the overall contribution of each energy source but do not effectively show changes over time.

Question 3

Question

An environmental monitoring agency has collected climate data from major Indian cities. Design a visualization that effectively represents Temperature, Air Quality Index (AQI), Rainfall, and Humidity using appropriate colour scales. Select a suitable colour palette and justify its effectiveness for accurate interpretation and accessibility.



R-Program:

Create Dataset

climate <- data.frame(

City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",

"Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",

"Bhopal","Patna","Goa","Kochi","Nagpur","Indore",

"Surat","Visakhapatnam","Coimbatore","Mysuru"),

Temperature=c(44,35,38,40,39,31,33,43,45,39,

```
38,40,32,30,41,37,36,34,29,28),

Humidity=c(22,84,78,72,48,66,58,26,18,42,
36,45,89,91,30,34,76,82,68,64),

Rainfall=c(6,280,220,190,85,120,95,2,1,35,
48,72,360,410,12,20,180,240,130,145),

AQI=c(365,110,125,180,160,72,84,290,320,240,
175,220,58,52,270,190,145,98,60,55)
)

# Assign Colors based on AQI
aqi.col <- ifelse(climate$AQI>300,"red",
      ifelse(climate$AQI>200,"orange",
      ifelse(climate$AQI>100,"yellow","green")))

# Bubble Plot
symbols(climate$Temperature,
     climate$Humidity,
     circles=climate$Rainfall,
     inches=0.25,
     bg=aqi.col,
     fg="black",
     xlab="Temperature (°C)",
```

```
    ylab="Humidity (%)",
    main="Climate Monitoring of Indian Cities")


# Add City Names
text(climate$Temperature,
   climate$Humidity,
   labels=climate$City,
   pos=3,
   cex=0.6)


# Legend
legend("topright",
    legend=c("Poor (>300)",
         "Moderate (201-300)",
         "Average (101-200)",
         "Good (<=100)"),
    fill=c("red","orange","yellow","green"),
    title="AQI")
```

Explanation:

The climate dataset containing Temperature, Humidity, Rainfall, and AQI for twenty Indian cities is created in R. A bubble plot is used to visualize the data. Temperature is plotted on the X-axis, Humidity on the Y-axis, Rainfall is represented by the bubble size, and AQI is shown using different colors. City names are displayed above each bubble, and a legend is included to explain the AQI color categories.

Justification

- **Temperature** → X-axis
- **Humidity** → Y-axis
- **Rainfall** → Bubble size
- **AQI** → Bubble color
- **City** → Labels

The selected color palette makes air quality easy to interpret:

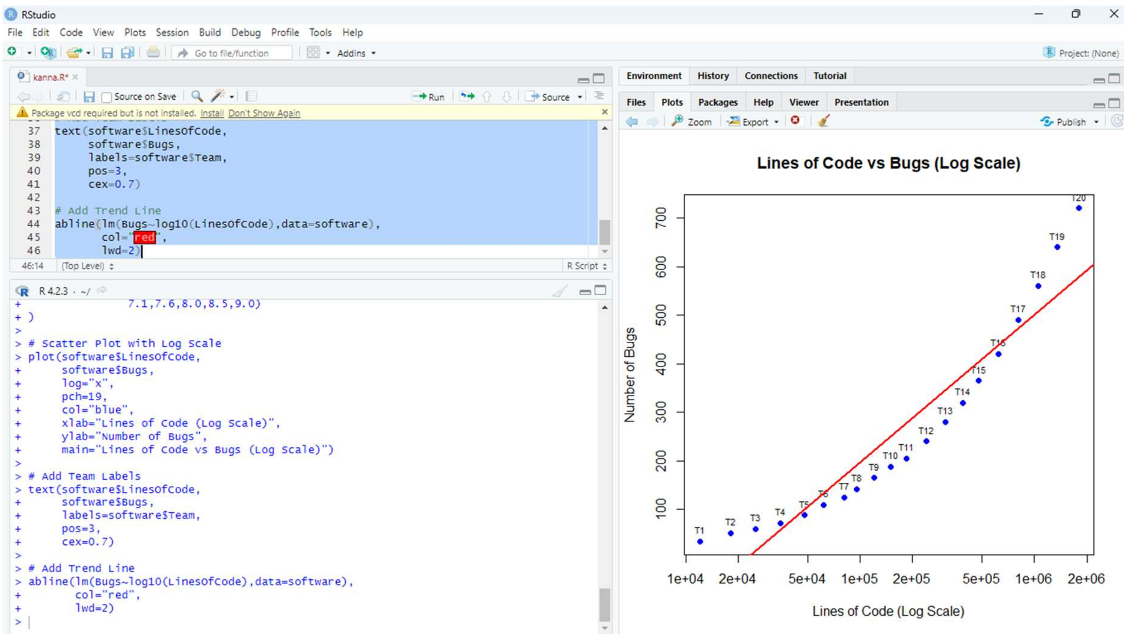
- **Green** – Good AQI
- **Yellow** – Average AQI
- **Orange** – Moderate AQI
- **Red** – Poor AQI

This visualization effectively displays all four variables in a single graph, making it easy to compare climatic conditions across different cities while ensuring clear interpretation and accessibility.

Question 4

Question:

A software company has recorded project statistics for twenty software development teams. Design a visualization that requires the use of a nonlinear axis transformation to represent highly skewed data. Justify the selected transformation and explain how it improves interpretation over a linear scale.



R-Program:

```
# Create Dataset

software <- data.frame(

Team=paste("T",1:20,sep=""),


LinesOfCode=c(12000,18000,25000,35000,48000,

62000,81000,96000,120000,150000,

185000,240000,310000,390000,480000,

620000,810000,1050000,1350000,1800000),
```

```
Bugs=c(35,52,60,72,88,
110,125,142,165,188,
205,240,280,320,365,
420,490,560,640,720),

Developers=c(5,7,9,10,12,
14,16,18,20,22,
25,28,30,34,38,
42,46,50,55,60),

Complexity=c(2.1,2.4,2.7,3.0,3.2,
3.5,3.8,4.1,4.5,4.8,
5.1,5.4,5.8,6.2,6.6,
7.1,7.6,8.0,8.5,9.0)
)

# Scatter Plot with Log Scale
plot(software$LinesOfCode,
    software$Bugs,
    log="x",
    pch=19,
    col="blue",
    xlab="Lines of Code (Log Scale)",
    ylab="Number of Bugs",
    main="Lines of Code vs Bugs (Log Scale)")
```

```
# Add Team Labels

text(software$LinesOfCode,

     software$Bugs,

     labels=software$Team,

     pos=3,

     cex=0.7)


# Add Trend Line

abline(lm(Bugs~log10(LinesOfCode),data=software),

       col="red",

       lwd=2)
```

Explanation:

The software dataset is created in R with Team, Lines of Code, Bugs, Developers, and Complexity. A scatter plot is drawn with **Lines of Code** on the X-axis and **Bugs** on the Y-axis. Since the Lines of Code values vary over a very large range, a **logarithmic scale** (`log="x"`) is applied to the X-axis. Team names are displayed above each point, and a trend line is added to show the relationship between Lines of Code and Bugs.

Why Nonlinear (Log) Scale?

- The Lines of Code values range from **12,000 to 1,800,000**, which is a very wide range.
- On a **linear scale**, smaller values get compressed near the origin and are difficult to compare.
- A **logarithmic scale** spreads the values more evenly, making patterns and relationships much easier to observe.

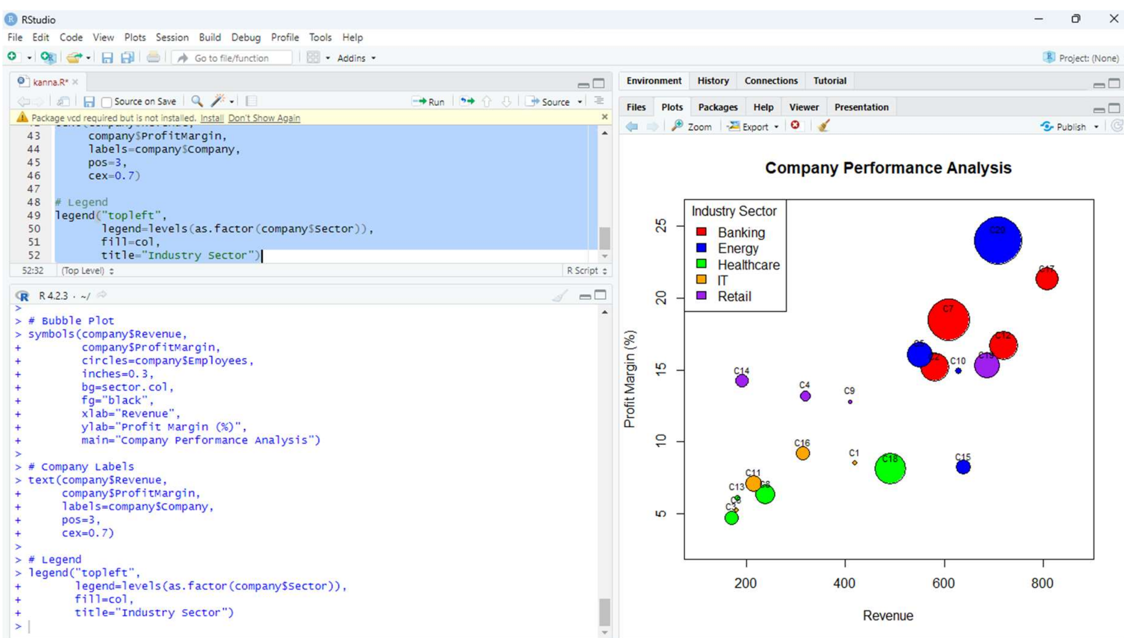
Conclusion

A **logarithmic (nonlinear) axis transformation** provides a clearer visualization than a linear scale because it effectively handles highly skewed data, improves readability, and makes trends between **Lines of Code** and **Bugs** easier to interpret.

Question 5

Question

A financial analytics company has collected performance data for twenty publicly listed companies. Develop an integrated visualization that effectively communicates Revenue, Profit Margin, Market Capitalization, Employee Strength, and Industry Sector. Use appropriate aesthetic mappings, coordinate systems, scales, and colour palettes to maximise interpretability while minimising perceptual bias.



R-Program:

company <- data.frame(

Company=paste("C",1:20,sep=""),

Sector=c("IT","Banking","Healthcare","Retail","Energy",

```
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy"),

Revenue=c(420,580,170,320,550,
180,610,240,410,630,
215,720,182,192,640,
315,808,492,686,710),

ProfitMargin=c(8.5,15.2,4.7,13.2,16.1,
5.2,18.5,6.3,12.8,14.9,
7.1,16.7,6.1,14.2,8.2,
9.2,21.3,8.1,15.3,24.0),

Employees=c(5000,32000,15000,12000,28000,
5100,46000,22000,4500,7000,
18000,32000,6000,14500,16000,
15200,25100,34000,28000,52000)
)

# Assign Colors for Sectors
col <- c("red","blue","green","orange","purple")
sector.col <- col[as.factor(company$Sector)]

# Bubble Plot
symbols(company$Revenue,
    company$ProfitMargin,
    circles=company$Employees,
    inches=0.3,
    bg=sector.col,
    fg="black",
```

```
    xlab="Revenue",
    ylab="Profit Margin (%)",
    main="Company Performance Analysis")

# Company Labels
text(company$Revenue,
   company$ProfitMargin,
   labels=company$Company,
   pos=3,
   cex=0.7)

# Legend
legend("topleft",
    legend=levels(as.factor(company$Sector)),
    fill=col,
    title="Industry Sector")
```

Explanation:

The company dataset is created in R with Company, Sector, Revenue, Profit Margin, and Employees. A bubble plot is used to display multiple variables in a single graph. Revenue is plotted on the X-axis, Profit Margin on the Y-axis, bubble size represents the number of employees, and different colors indicate the industry sector. Company names are displayed above each bubble, and a legend is added to identify the sectors.